Particle Physics Homework 7

1). Find the interacting Hamiltonian in the interaction picture ($H_I = e^{iH_0t} H_{int} e^{-iH_0t}$) of the Lagrangian

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{3!}\phi^3$$

2). Show that

$$\langle\Omega| = \lim_{T\to\infty(1-i\varepsilon)} \langle 0|\hat{\mathcal{U}}(T,t_0)\left(e^{-iE_0(t_0-(-T))}\langle\Omega|0\rangle\right)^{-1}$$

3). Calculate the following quantities for

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4 \qquad (\phi^4\text{- theory})$$

up to the term with λ^2

3.1) $\langle\Omega|\hat{T}(\phi(x^\mu))|\Omega\rangle$

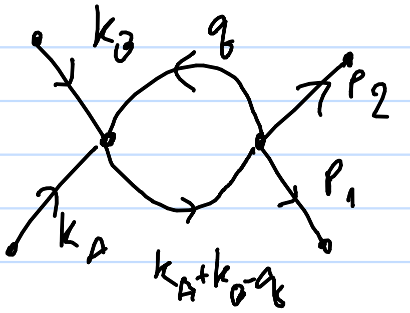
3.2) $\langle\Omega|\hat{T}(\phi(x_1^\mu)\phi(x_2^\mu))|\Omega\rangle$

3.3) $\langle\Omega|\hat{T}(\phi(x_1^\mu)\phi(x_2^\mu)\phi(x_3^\mu))|\Omega\rangle$

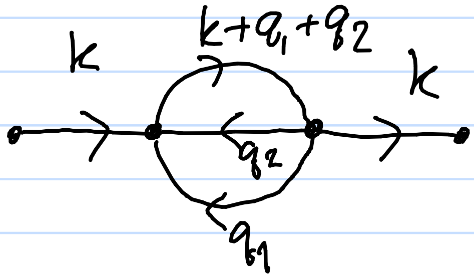
3.4) $\langle\Omega|\hat{T}(\phi(x_1^\mu)\phi(x_2^\mu)\phi(x_3^\mu)\phi(x_4^\mu))|\Omega\rangle$

4). Write the expression for the following diagrams in ϕ^4 theory (external lines are physical particles)

4.1)



4.2)



5). ϕ^3- theory

Consider the Lagrangian

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{3!}\phi^3$$

5.1) What is the dimension of $[\lambda]$

5.2). Find $\hat{H}_I(t)$

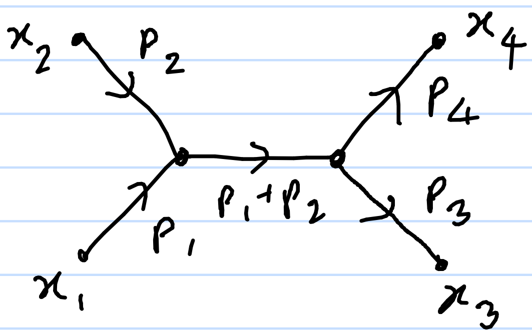
5.3). Find the following:

5.3.1). $\langle\Omega|\hat{T}(\hat{\phi}_I(x_1)\hat{\phi}_I(x_2)\hat{\phi}_I(x_3))|\Omega\rangle$

5.3.2). $\langle\Omega|\hat{T}(\hat{\phi}_I(x_1)\hat{\phi}_I(x_2)\hat{\phi}_I(x_3)\hat{\phi}_I(x_4))|\Omega\rangle$

5.4). Write the expression for the following diagrams (external lines are physical particles)

5.4.1)



5.4.2)